Emergency Department Simulation Model and Analysis

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Abstract

This report presents a Discrete Event Simulation (DES) model of an Emergency Department (ED) developed using AnyLogic. The model investigates patient arrivals, triage procedures, severity-based medical priorities, resource allocation, and probabilistic diagnostic testing. Patient arrivals follow a Non-Homogeneous Poisson Process, while medical examinations support preemption, allowing higher-priority patients to interrupt lower-priority treatments. A factorial experimental design evaluates 27 configurations of doctors, nurses, and beds, with 100 independent replications configured for each scenario. The main performance indicators include waiting times before medical examination, waiting times by severity code, and resource utilization. The results identify promising resource configurations and highlight the effects of resource congestion on patient flow.

1 Introduction and System Description

Effective resource coordination in emergency healthcare settings is essential to reduce waiting times and ensure timely medical intervention. This project investigates the behavior of an Emergency Department through a Discrete Event Simulation (DES) model developed using AnyLogic.
Patients arrive according to a time-dependent stochastic process, undergo triage, and are assigned one of four severity codes. After triage, they wait for an available bed and medical personnel. Medical examinations are scheduled according to severity-based priorities, allowing urgent cases to interrupt ongoing lower-priority visits. Following examination, patients either leave the system directly or undergo additional diagnostic tests before exiting.

1.1 Simulation Paradigm

The model adopts the **Discrete Event Simulation (DES)** paradigm, in which system states change at specific events, such as patient arrivals, resource allocation, the beginning or completion of examinations, and the suspension or resumption of medical tasks.
This approach allows the simulation to represent patient queues, limited resources, stochastic service durations, and priority-based interactions without explicitly simulating every instant of time.

1.2 Justification of the Approach

The combination of non-stationary arrivals, limited resources, severity-based priorities, medical preemption, and probabilistic routing makes the system difficult to analyze using standard analytical queueing models.

Simulation provides a flexible approach for investigating how changes in the number of doctors, nurses, and beds influence patient waiting times and resource utilization under stochastic conditions.

1.3 Modeling Process and Assumptions

The model incorporates the following assumptions:

- **Patient Arrivals:** Arrivals follow a Non-Homogeneous Poisson Process, with an arrival intensity that changes throughout the day according to a predefined schedule.
- **Severity Classification:** At the end of triage, each patient is assigned a severity code between 1 and 4 using a discrete uniform distribution. Each code has an equal probability of 25%, with code 1 representing the highest priority and code 4 the lowest.
- **Service Times:** Triage duration follows an Exponential distribution with a mean of 5 minutes. Medical examination duration follows a Normal distribution with a mean of 20 minutes and a standard deviation of 5 minutes, with a minimum duration of 1 minute. Additional diagnostic tests are modeled using a stochastic delay.
- **Resource Constraints:** Nurses, doctors, and beds are represented as limited resource pools.
- **Priority and Preemption:** Medical examinations follow severity-based priorities. A more urgent patient may interrupt an ongoing lower-priority examination, which can subsequently resume.
- **Probabilistic Routing:** After examination, patients may require additional diagnostic tests according to a Bernoulli decision.
- **Simulation Horizon:** Each simulation run represents 24 hours of Emergency Department operation.

2 Model Structure and Flowchart Implementation

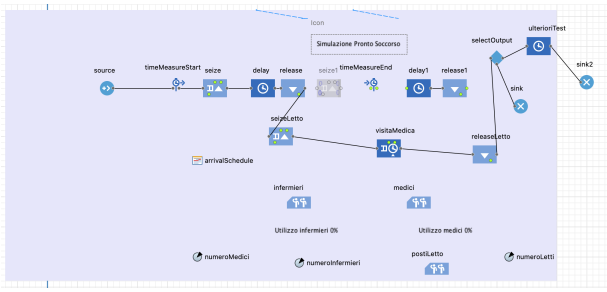


Figure 1: AnyLogic flowchart of the Emergency Department, including patient arrivals, triage, bed allocation, medical examinations, and optional diagnostic tests

The model was implemented using the AnyLogic Process Modeling Library. The patient flow consists of the following stages:

- **Source:** Generates incoming patients according to the time-dependent arrival schedule.
- **Triage:** Patients acquire a nurse through a Seize block, undergo a triage Delay, and release the nurse after completing triage.

- **Bed Allocation:** Patients request an available bed through the `seizeLetto` block. Priority values depend on their assigned severity code.
- **Medical Examination:** The `visitaMedica` Service block allocates a doctor and models the duration of medical treatment. Preemption is enabled so that higher-priority medical tasks may interrupt lower-priority ones.
- **Bed Release:** The `releaseLetto` block releases the occupied bed after medical examination.
- **Probabilistic Routing:** The `selectOutput` block determines whether additional diagnostic tests are necessary.
- **Additional Tests:** Patients requiring further examinations pass through the `ulterioriTest` Delay block.
- **Sink Blocks:** The `sink` and `sink2` blocks represent patients completing their pathways, respectively without and with additional tests.

2.1 Severity-Based Priority and Preemption

One of the main features of the model is the implementation of preemptive medical examinations. Patients are assigned a task priority based on their severity code, using the expression:

$$P = 5 - \text{severityCode} \tag{1}$$

Therefore, code 1 patients have the highest numerical priority, while code 4 patients have the lowest.
When a higher-priority medical task requires an occupied medical resource, the Service block can suspend a lower-priority task. The interrupted visit may later resume according to the resource preemption policy.
This behavior was tested using AnyLogic event callbacks. The simulation Console confirmed that medical examinations were both suspended and resumed during execution.

2.2 Data Collection

The model collects statistics directly through AnyLogic data objects and resource-pool measurements.
The waiting time before medical examination is recorded when a patient begins the medical service. The corresponding statistics are also divided according to the four severity codes.
Doctor and nurse utilization is measured through their resource pools, while the numbers of patients reaching the two Sink blocks are used to evaluate completed patient pathways and the proportion requiring additional tests.

3 Key System Parameters (KSPs) and Key Performance Indicators (KPIs)

The simulation distinguishes controllable system parameters from performance indicators measured during execution.

3.1 Key System Parameters (KSPs)

The experimental analysis varies three principal parameters:

- **Number of Doctors:** 2, 3, or 4 available doctors.
- **Number of Nurses:** 2, 3, or 4 available nurses.
- **Number of Beds:** 6, 8, or 10 available treatment beds.

Arrival schedules, severity assignment rules, service-time distributions, and probabilistic routing remain part of the model's fixed assumptions during these experiments.

3.2 Key Performance Indicators (KPIs)

The main KPIs considered in the model are:

- **Mean Waiting Time Before Medical Examination:** Average time elapsed between patient arrival and the beginning of the medical visit.
- **Waiting Time by Severity Code:** Mean waiting time measured separately for patients with severity codes 1, 2, 3, and 4.
- **Doctor Utilization:** Fraction of simulation time during which medical resources are occupied.
- **Nurse Utilization:** Fraction of simulation time during which nursing resources are occupied.
- **Additional Test Rate:** Proportion of patients completing their pathways through the additional testing branch.
- **Completed Patients:** Total number of patients reaching the two Sink blocks during the simulation horizon.

4 Experimental Setup

A factorial experimental design was implemented using AnyLogic's Parameter Variation experiment.
The experiment evaluates three values for each of the three controllable parameters, producing:

$$3 \times 3 \times 3 = 27 \text{ configurations.} \tag{2}$$

Each simulation represents 24 hours of operation, corresponding to 1440 simulation minutes.
The experiment was configured to execute 100 independent replications for each configuration, corresponding to 2700 simulation runs in the complete factorial design.

4.1 Statistical Precision

For each configuration, the mean waiting time is estimated from independent simulation replications. Confidence intervals are calculated at the 95% confidence level using Student's t distribution:

$$CI_{95\%} = \overline{x} \pm t_{0.975,N-1} \frac{s}{\sqrt{N}} \tag{3}$$

where $\overline{x}$ is the sample mean, s is the sample standard deviation, and N is the number of available replications.
The relative error is defined as the confidence-interval half-width divided by the estimated mean:

$$\varepsilon = \frac{t_{0.975,N-1}s}{\sqrt{N}\,\overline{x}}. \tag{4}$$

The target relative precision is 5% at the 95% confidence level. Increasing the number of replications reduces statistical uncertainty, although achieving a particular precision is not automatically guaranteed.

5 Results and Graphical Analysis

The simulation experiments provide information about patient waiting times, resource utilization, and the effects of different staffing and bed configurations.

5.1 Resource Utilization

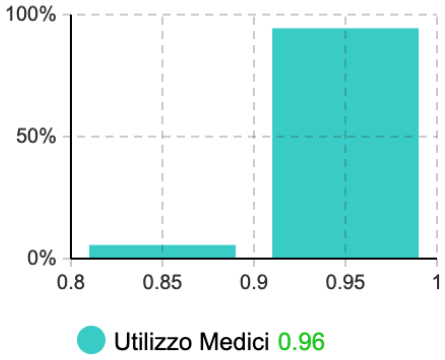


Figure 2: Distribution of doctor utilization across the recorded experimental replications

Doctor utilization is an important indicator of system congestion. The simulations show that medical resources can operate close to full capacity, particularly under configurations with fewer doctors. High utilization limits the ability of the system to absorb additional demand and can contribute to longer queues.

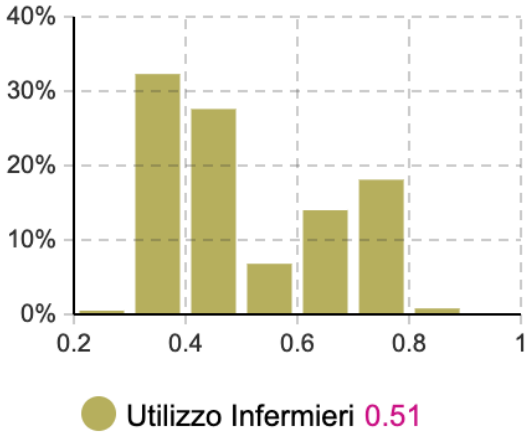


Figure 3: Distribution of nurse utilization across the recorded experimental replications

Nurse utilization reflects the workload generated by the triage stage. Differences between doctor and nurse utilization indicate that the two resource pools may face different levels of congestion. Consequently, increasing one resource type does not necessarily produce an equivalent improvement in the overall patient flow.
Across the 2,700 experimental replications, the average doctor utilization was 96.4%, compared with 51.2% for nurses. These results indicate that medical resources experience substantially higher utilization than nursing resources, suggesting that doctor availability represents a major bottleneck in the modeled system. These averages are aggregated across all 27 resource configurations.

5.2 Waiting Times

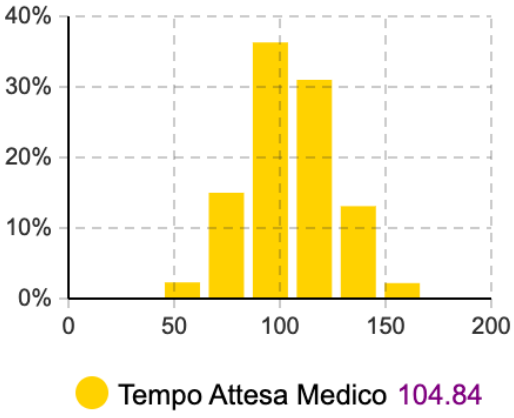


Figure 4: Distribution of mean waiting time from patient arrival to the beginning of medical examination

Waiting times depend on patient arrivals, triage operations, bed availability, medical staffing, and priority scheduling.
Severity-based preemption allows urgent medical tasks to interrupt lower-priority examinations. However, total waiting times remain influenced by resource availability, particularly when patients cannot obtain a treatment bed.

Across the 2,700 simulation runs, the average of the mean waiting times before medical examination was 104.84 minutes. This value represents the overall average across all 27 resource configurations and should not be confused with the lowest observed mean waiting time of 91.26 minutes, obtained with 4 doctors, 3 nurses, and 10 beds.

5.2.1 Waiting Times by Severity Code

Waiting times were also analyzed separately for the four severity codes. The following results refer to a single 24-hour simulation with 4 doctors, 3 nurses, and 10 beds.

Severity Code	Patients Starting Examination	Mean Waiting Time (min)
1	99	15.41
2	98	59.32
3	49	231.96
4	21	26.37

Table 1: Waiting times by severity code in a single simulation run.

Code 1 patients experienced the shortest average waiting time, consistent with their highest medical priority. Code 3 patients experienced substantially longer waiting times.
Although code 4 patients showed a relatively low average waiting time, only 21 patients of this category started medical examination during the simulation. Since patients still waiting at the end of the 24-hour horizon are excluded from the calculation, this result does not necessarily indicate better service for lower-priority patients.
Overall, 414 patients arrived during this simulation, 267 started medical examination, and 147 remained waiting for a bed at the end of the simulation.

5.3 Additional Diagnostic Tests

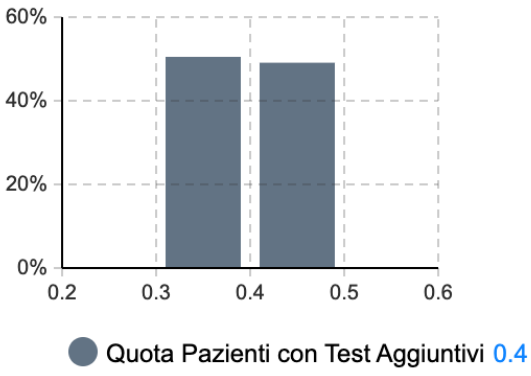


Figure 5: Distribution of the proportion of patients requiring additional diagnostic tests.

The probabilistic routing mechanism separates patients completing their medical pathway directly from those requiring additional diagnostic tests.
The additional diagnostic test rate was evaluated across 2,700 simulation replications, covering all 27 resource configurations. The average proportion of patients requiring additional tests was 39.9%, with observed values ranging from 27.9% to 56.7%.
This indicator represents the proportion of patients who complete their medical pathway through the additional testing branch. It does not measure patient abandonment, since queue abandonment due to excessive waiting time is not implemented in the current model.

The corresponding indicator is calculated as:

$$R_{\text{tests}} = \frac{N_{\text{sink2}}}{N_{\text{sink}} + N_{\text{sink2}}} \tag{5}$$

where N_{sink} represents patients leaving directly and N_{sink2} represents patients completing the additional testing pathway.

5.4 Comparison of Resource Configurations

The experimental results were grouped according to the numbers of doctors, nurses, and beds. Table 2 presents the results for all 27 resource configurations, based on 100 independent simulation replications per scenario.

Doctors	Nurses	Beds	Mean Wait (min)	Relative Error
2	2	6	112.07	3.54%
2	2	8	110.54	3.92%
2	2	10	113.34	3.24%
2	3	6	110.14	4.03%
2	3	8	112.73	3.78%
2	3	10	108.12	3.68%
2	4	6	109.31	4.15%
2	4	8	113.52	4.09%
2	4	10	114.66	3.79%
3	2	6	110.57	3.45%
3	2	8	111.88	3.58%
3	2	10	109.35	2.95%
3	3	6	105.25	3.95%
3	3	8	108.14	3.49%
3	3	10	106.94	3.97%
3	4	6	105.33	3.93%
3	4	8	107.57	3.80%
3	4	10	110.21	3.10%
4	2	6	96.79	3.03%
4	2	8	96.90	3.20%
4	2	10	95.94	2.98%
4	3	6	96.61	2.99%
4	3	8	95.09	3.21%
4	**3**	**10**	**91.26**	**3.39%**
4	4	6	97.23	3.30%
4	4	8	95.62	3.58%
4	4	10	93.68	2.87%

Table 2: Comparison of all 27 resource configurations, based on 100 independent replications per scenario. Relative errors are computed at the 95% confidence level.

The experimental analysis was completed across all 27 resource configurations, with 100 independent replications per scenario, resulting in 2,700 simulation runs.
Among the evaluated configurations, the lowest observed mean waiting time was obtained with 4 doctors, 3 nurses, and 10 beds:

$$\overline{x} = 91.26 \text{ minutes.} \tag{6}$$

The corresponding 95% confidence interval is:

$$CI_{95\%} = [88.17,\ 94.36] \text{ minutes.} \tag{7}$$

The relative error is 3.39%, satisfying the target precision of 5%. All 27 configurations achieved a relative error below 5% for the mean waiting time.
Compared with the baseline configuration of 2 doctors, 2 nurses, and 6 beds, which recorded a mean waiting time of 112.07 minutes, the best observed configuration reduced the average waiting time by approximately 20.81 minutes, corresponding to an 18.6% improvement.
The results suggest that increasing the number of doctors generally improves waiting-time performance, while increasing nursing staff or bed capacity does not always produce equivalent improvements. Nevertheless, the lowest observed mean does not necessarily imply statistically significant superiority over all alternative configurations.

6 Discussion and Conclusions

This project demonstrates how Discrete Event Simulation can be used to analyze an Emergency Department under non-stationary patient arrivals, stochastic service durations, limited resources, and severity-based medical priorities.
One of the main features of the model is the implementation of preemptive medical examinations. Simulation tests confirmed that lower-priority visits can be suspended and subsequently resumed when more urgent patients require medical attention.
The experimental analysis evaluated 27 resource configurations, with 100 independent replications per scenario, resulting in 2,700 simulation runs. All configurations achieved a relative error below 5% for the mean waiting time, with a 95% confidence level.
The lowest observed mean waiting time was 91.26 minutes, obtained with 4 doctors, 3 nurses, and 10 beds. The corresponding 95% confidence interval was [88.17, 94.36] minutes, with a relative error of 3.39%.
Compared with the baseline configuration of 2 doctors, 2 nurses, and 6 beds, the average waiting time decreased by approximately 18.6%. The results suggest that increasing medical staffing has a particularly important effect on reducing patient waiting times.
Across all experimental replications, average doctor utilization was 96.4%, while average nurse utilization was 51.2%. These findings suggest that doctors represent a major bottleneck in the modeled system.
Nevertheless, the model has some limitations. Waiting-time statistics include only patients who begin medical examination within the 24-hour simulation horizon, excluding those still waiting at the end of the simulation. Furthermore, patient abandonment due to excessive waiting time is not implemented.
Future extensions could investigate more realistic severity distributions, patient abandonment, alternative staffing schedules, and improved bed allocation strategies.